

SIBD Project - Part 1

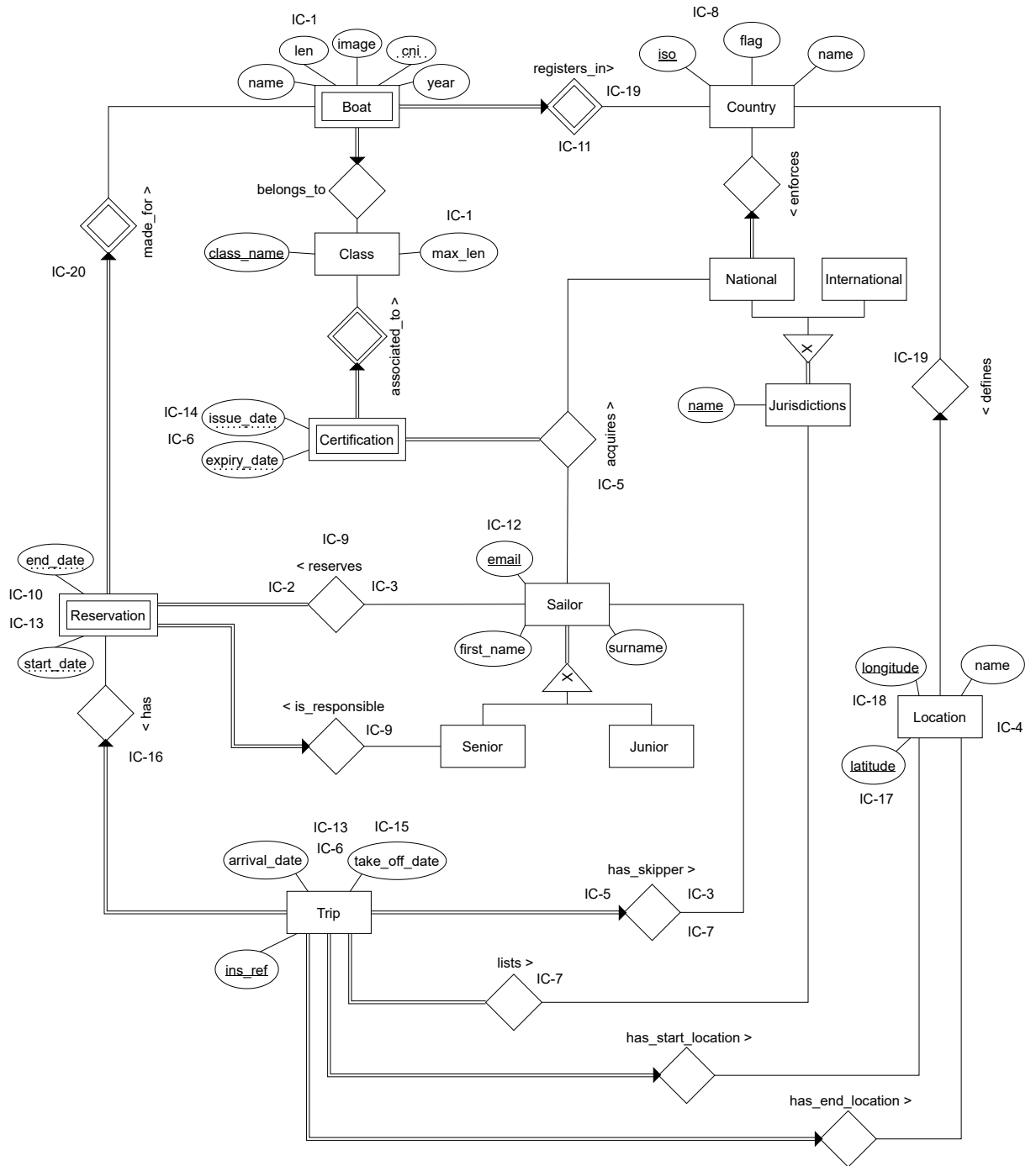
Group 10

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December 2025

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IC-1 : The boat len must be $\leq$ max_len and both len and max_len should be non-negative
 IC-2 : A reservation must not consist of only Junior sailors
 IC-3 : Skipper must be one of the sailors from the reservation
 IC-4 : Any two locations must be at least one nautical mile apart
 IC-5 : A skipper must have a certification for the jurisdiction and the boat class that has been reserved
 IC-6 : The certificate held by the skipper should have issue_date < take_off_date and expiry_date > arrival_date of the trip
 IC-7 : The skipper should have valid certifications for all jurisdictions listed in the trip
 IC-8 : name and flag of country are unique
 IC-9 : The responsible senior sailor must be one of the sailors who made a joint registration
 IC-10 : end_date must be > than start_date

IC-11 : cni is assigned by the country where the boat is registered
 IC-12 : email follows basic email format (like '%@%.%')
 IC-13 : trip dates must fall within reservation date, i.e. arrival_date $\geq$ start_date and trip_date $\leq$ end_date
 IC-14 : Certificate issue_date < expiry_date
 IC-15 : trip take_off_date < arrival_date
 IC-16 : Any 2 trips' active time windows should not overlap
 IC-17 : latitude values must be between -90.000000 and +90.000000
 IC-18 : longitude values must be between -180.000000 and +180.000000
 IC-19 : Any country that registers boats must have at least one location defined
 IC-20 : Reservations for the same boat must not have overlapping start_date and end_date intervals

Figure 1: EA Diagram